

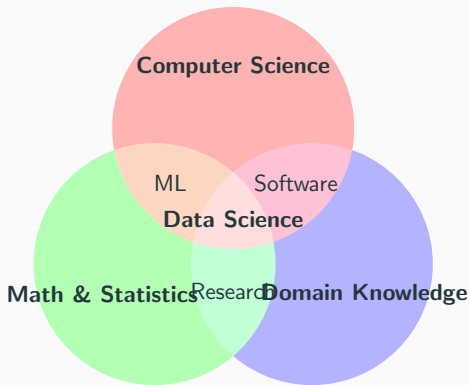
Introduction to Data Science

The Data Science Landscape

Dept. CSE

Week 2: Class 1

What is Data Science?



Definition

Data Science is an interdisciplinary field that uses scientific methods, processes, and systems to extract knowledge and insights from noisy, structured, and unstructured data.

Evolution: From Statistics to Data Science

Feature	Traditional Stats (STA 101)	Data Science (CSE perspective)
Data Size	Small, clean samples	Massive, messy, "Big Data"
Goal	Inference (Why did it happen?)	Prediction (What will happen?)
Format	Mostly Tabular (Numbers)	Text, Images, Video, Logs
Assumption	Normal distribution, independence	Dealing with complex, non-linear patterns

The 5 V's of Big Data

To a CSE student, "Data" isn't just a table; it's a resource with these properties:

1. **Volume:** The sheer scale (Terabytes to Petabytes).
2. **Velocity:** The speed of data generation (Streaming data vs. Static).
3. **Variety:** Different types (Structured, Unstructured, Semi-structured).
4. **Veracity:** The messiness/uncertainty of data (Missing values, noise).
5. **Value:** The ability to turn data into business/social impact.

The Data Science Pipeline (Preview)

1. **Data Collection:** APIs, Databases, Sensors.
2. **Data Cleaning:** Handling the "Veracity" problem.
3. **Analysis:** Applying the Math you learned in STA 101.
4. **Deployment:** Integrating the result into a CSE system (App/Web).

Case Study: The Smart Traffic System

A city installs sensors at 1,000 intersections. These sensors send image data to a central server every second to predict traffic jams.

Problem:

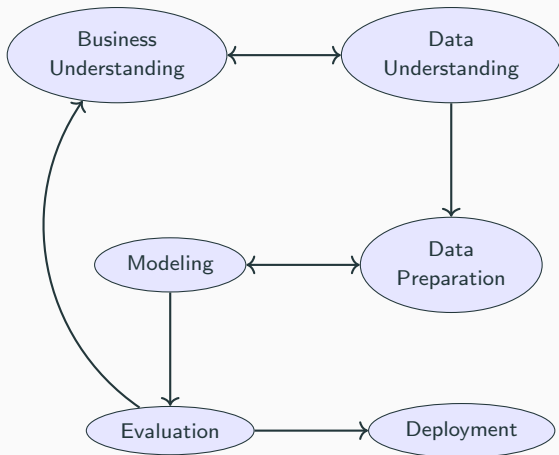
1. Identify which of the **5 V's** is the biggest technical challenge here. Justify your choice.
2. Contrast this with a traditional statistical survey (e.g., asking 50 drivers about their commute). Why can't we use simple "Mean" or "Median" to solve this smart traffic problem directly?

Why do we need a Process?

In CSE, we have the *Software Development Life Cycle* (SDLC). In Data Science, we need a lifecycle because:

- Data is messy and unpredictable.
- 80% of projects fail due to poor planning, not bad algorithms.
- It ensures reproducibility and stakeholder alignment.

The CRISP-DM Model



- **Iterative:** Notice the arrows go back and forth.
- **Cyclical:** The deployment usually triggers new business questions.

The OSEMN Framework (The Engineering View)

Popularized by Hilary Mason, OSEMN is more focused on the technical pipeline:

1. **O**btain: Gathering data (APIs, Scraping, SQL).
2. **S**crub: Cleaning (Handling missing values/outliers).
3. **E**xplore: EDA (The STA 101 part: Mean, Median, Correlation).
4. **M**odel: Applying logic/algorithms.
5. **i**Nterpret: Visualizing and telling the story.

Defining the Problem: Classifying Tasks

A key part of Business Understanding is defining the **Data Task**:

- **Classification:** Is this email spam? (Discrete labels)
- **Regression:** What will the price be? (Continuous values)
- **Clustering:** Which customers are similar? (Grouping)
- **Anomaly Detection:** Is this transaction fraudulent?
(Outliers)

Note

Note how this links to STA 101. Regression in Statistics is a "Model" in DS.

The Data Understanding Gap

During Data Understanding, we check the **Data-Problem Fit**:

- Do we have the right features?
- Is the sample size sufficient for the variance (STA 101)?
- **Example:** Predicting student grades using only their "Height" and "Eye Color" will fail Business Understanding because there is no logical correlation.

Assessment (C3 - Application)

Scenario: University Attendance System

The University wants to predict which students are likely to drop out of a course mid-semester using WiFi log data (showing how often they come to campus).

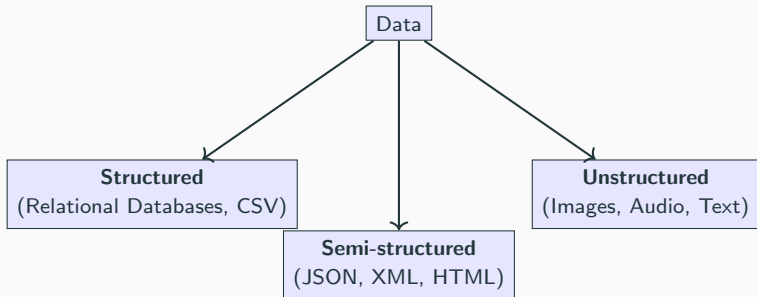
Task: Map this project onto the **OSEMN** framework. List one specific activity for each letter:

- O: _____
- S: _____
- E: _____
- M: _____
- N: _____

Problem: During the **Evaluation** phase of a project to detect credit card fraud, the DS team finds that the model is 99% accurate but it *misses* 50% of actual fraud cases.

1. Analyze why this project should go back to the **Business Understanding** phase instead of the **Deployment** phase.
2. Using your STA 101 knowledge (Type I and Type II errors), explain why "99% Accuracy" might be a misleading metric in this specific context.

The Data Taxonomy



- **Structured:** High organization (think Excel/SQL).
- **Unstructured:** No pre-defined model (80% of enterprise data).
- **Semi-structured:** Contains tags or markers to separate elements.

CSV (Comma Separated Values)

- Simple, flat structure.
- Low overhead.
- Hard to represent nested data (e.g., a person with multiple phone numbers).

JSON (JavaScript Object Notation)

- Key-Value pairs.
- Hierarchical/Nested.
- The standard for modern Web APIs.

The Logic of JSON

Theoretical Structure

```
{  
  "student_id": 101,  
  "grades": [90, 85, 88],  
  "address": { "city": "Dhaka", "zip": 1207 }  
}
```

Why for Data Science?

- Flexible: If a new student has a "Middle Name," we just add the key without breaking the "Schema."
- Contrast with STA 101: In a stats table, a missing middle name would be a "Null" value in a rigid column.

Data Sources: Where does it come from?

1. **User-Generated:** Social media, surveys, transaction logs.
2. **Machine-Generated:** Server logs, IoT sensor readings.
3. **Internal vs. External:**
 - **Internal:** Company sales records.
 - **External:** Open Data (e.g., Kaggle, World Bank), Scraping.

Data Provenance

In Data Science, we must track the *lineage* of data. If the source is biased, the math will be biased.

Assessment (Bloom's C3 - Application)

Task: Data Representation

A Library Management System records:

- Book Title: "Data Science Theory"
- Authors: "John Smith" and "Alice Wong"
- ISBN: 987654321

Problem:

1. Represent this data in **JSON** format. Ensure you handle the multiple authors correctly (Hint: use an array).
2. If you had to store this in a flat **CSV** file, what problems would you face regarding the "Authors" column?

Assessment (Bloom's C4 - Analysis)

Case Study: Smart Watch Data A smart watch records heart rate (every second), GPS location (every minute), and the user's name/age.

1. Analyze which of these data points is **Structured** and which is **Semi-structured**.
2. If you are building a system to predict a heart attack, why is "Velocity" (from Class 1) more critical than "Volume" for this specific data source?
3. **Math Logic:** If a heart rate sensor fails for 10 seconds, and you use the "Last Known Value" to fill the gap, what statistical property are you maintaining, and what are you risking?